Generative AI in Higher Art Education

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***Abstract*—This research delves into the perspectives of Chinese university art teachers on the integration of Artificial Intelligence in Generative Content. Collaboratively initiated by art teachers from Chinese universities through the AI Art Education Alliance in Wuhan, the study aims to comprehend their attitudes, concerns, and preparations regarding the infusion of generative AI tools into art and design curricula. The research employs a comprehensive approach, incorporating group interviews during the inaugural session of the AI Art Education Alliance. Additionally, questionnaire research is utilized as supplementary evidence. Participants include middle-level administrators and teachers from diverse colleges, offering a nuanced understanding of viewpoints across various educational institutions. Key findings reveal nuanced perspectives on AI in higher art education. Notably, there exists a spectrum of AI anxiety, with comprehensive universities showing readiness, while caution prevails in art colleges. The study underscores the potential benefits of AI in art education but highlights concerns about its impact on traditional pedagogy. The research emphasizes the urgency of addressing equity issues related to resource disparities, academic integrity, and cultural resistance within the education community. Recommendations include standardized AI tool usage, adaptations in professional structures, and fostering collaborative alliances to harness AI's potential effectively.**

Keywords—generative AI tools, higher education, teachers' perspectives, AI in art education, AI anxiety

I. Introduction

In recent years, there has been renewed interest in introducing AI to non-computer science university students[1], and despite the profound impact of artificial intelligence on higher education, there is currently a paucity of research on its application in university art education[2][3]. This gap is particularly notable in studies focused on the community of art educators in universities, and it aligns with contemporary initiatives addressing equity in learning analytics[4][5][6].

In an effort to delve deeper into the perspectives of university art teachers on AIGC and discuss strategies for art education, there has been active participation from art teachers at Chinese universities. This collaborative endeavor has resulted in the formation of the Chinese Artificial Intelligence Art Education(AIAED) Alliance established on 20 August 2023(Fig. 1). Through the collection of group interview data from the inaugural working session of the alliance, this research thoroughly investigates teachers' opinions regarding AIGC technology. Educators' perspectives hold significance, serving as conduits to instigate transformative shifts in higher education and facilitating the practical implementation of policies[7]. This study primarily focuses on four questions: What specific concerns do teachers have when faced with generative AI tools? How do university art teachers perceive the equity issues arising from the use of generative AI tools in education? Will AI replace human teachers, and what impact does this question have on teachers? How to meet the unique challenges that university art teachers face in the era of AI? Unlike previous work[8][9], this study specifically targets higher art education.

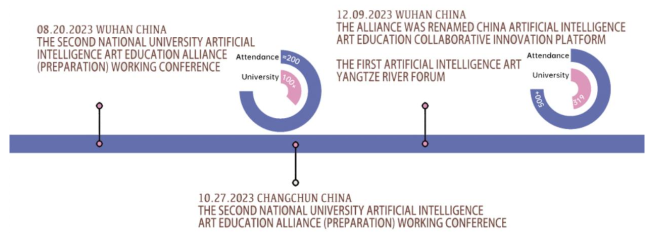

Fig. 1. Timeline of AIAED alliance.

As the first study to collect the collective voices of university art teachers through the AIAED alliance, this paper provides a unique perspective and valuable insights for future collaborative endeavors. The research aims to contribute to the academic community and educational practitioners by enhancing understanding of the application of AI in higher education art instruction. It aspires to inspire further research and collaboration at the intersection of AI and art education.

II. Literature Review

There is limited research on university teachers' attitudes towards AI tools, prompting an exploration of attitudes from three perspectives: students, policymakers, and teachers.

Regarding students, studies reveal a generally positive reception of AI-powered writing tools, acknowledging benefits in grammar checks, plagiarism detection, and language translation, but concerns about creativity and ethical writing practices persist[80]. Factors influencing students' acceptance of AI tools include familiarity with AI technologies and perceived usefulness, with usability also playing a significant role[25][26]. Additionally, research has delved into students' perceptions of AI tools like ChatGPT and their impact on anxiety levels in e-learning environments[27][28][29].

Policymakers express concerns about the digital divide in AI-enhanced education and propose accessible AI-based educational solutions[31]. Evaluations of policy-making processes stress the importance of considering public values and involving students[30]. Furthermore, effective AIED policies are identified as contextual, consultative, dynamic, and

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implementable and measurable, with a suggested seven-step approach for policy development[39].

Teachers, crucial in AI integration, may face challenges due to a lack of readiness and anxiety towards adopting new technologies[8][32][33][34][35][36]. Despite generally positive attitudes towards AI tools among teachers, readiness levels vary across contexts[37][38], with the frequency of AI tool usage correlating positively with teachers' perspectives[42]. Confidence in personal ability to use AI and perception of advantages significantly impact willingness to use AI in education[41], although ethical concerns persist[43].

In conclusion, while teachers are pivotal in AI integration, research on their viewpoints remains limited compared to students and administrators[44]. Although there has been some accumulation of research on teachers' attitudes in recent years, existing literature predominantly focuses on STEM fields and K-12 education, with fewer studies in higher education, particularly in art education.

III. Research Method

This research draws data from two research methods: group interviews and a questionnaire survey, and the primary research object of this study was college art teacher group.

A. Group Interview

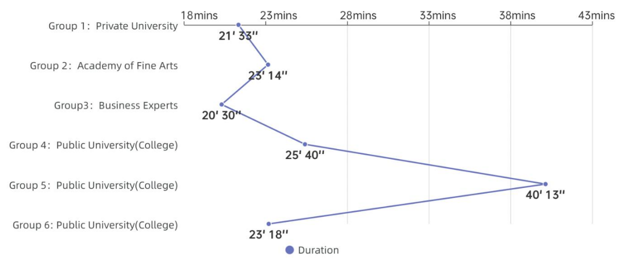

Fig. 2. Group interview overview.

The data for the group interviews were collected during the AIAED Alliance's first conference held in Wuhan, China in August 2023[10]. There were around 200 college representatives who attended this meeting. They are from more than 100 colleges or universities locating in most parts of China.

The group interviews included teachers from different regions, categorized into private universities, art colleges, enterprises, and public universities. Each group comprised five teachers, with one serving as the host for each interview. The duration of each interview ranged from 20 to 40 minutes. 30 of the representatives participated in the group interviews and the majority of them were from comprehensive universities.

B. Questionnaire Design

1) Introduction

This study utilizes a questionnaire to explore the perspectives, benefits, and challenges encountered by art college teachers in adopting generative AI tools. The questionnaire includes items designed using Likert scales and was initially pre-filled by three teachers specializing in animation and digital media arts. Based on their feedback, necessary adjustments were made. Subsequently, the questionnaire was distributed through the online survey platform, Questionnaire Star, with the support of the AI Art Education Alliance. Target participants included members of the AI Art Education Alliance among Chinese university teachers. Data collection took place in September 2023 over a one-week period.

Upon receiving questionnaire responses, a completeness check was conducted to ensure each questionnaire was adequately filled out. Incomplete responses were excluded from the analysis. Additionally, the collected data underwent validation to identify and eliminate any inconsistent or anomalous data. Completeness checks were performed on the 108 received responses, excluding 18 incomplete questionnaires. The remaining 90 completed questionnaires underwent validation to ensure data consistency. This study focuses on relevant content related to the main theme, emphasizing a streamlined approach without compromising the original information.The KMO value is 0.968, which means the data can be effectively extracted information. By analyzing α coefficient, the value of the reliability coefficient is 0.985, indicating that the reliability quality of the research data is high.

2) Research object analysis

Among the surveyed university teachers, those specializing in animation and digital media account for 74.74%, representing an overwhelming majority(Fig. 3). This indicates that teachers in animation and digital media-related disciplines are more actively involved in and exhibit a stronger sense of crisis toward Artificial Intelligence in Generative Content (AIGC) compared to other art disciplines. It also suggests disparities in the integration of AI within various art disciplines, with lower attention to AIGC observed in art and design disciplines outside of digital media and animation.

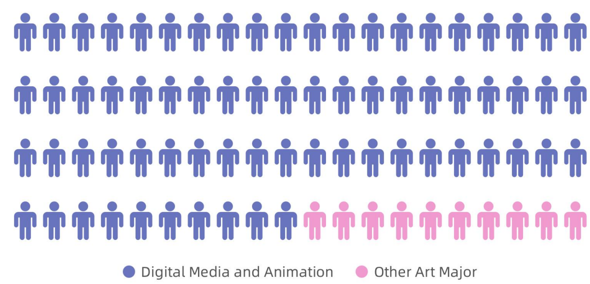

Fig. 3. Subject distribution of respondents.

From the perspective of geographical distribution, the surveyed teachers are distributed across 16 provinces in China, with a predominant presence in the eastern and central regions of the country. In addition, Approximately 54% of respondents are from private universities, with the rest from public universities.

IV. Who Is More Anxious about AI

The phrase 'AI anxiety(AIA)' refers to feelings of fear or agitation about out-of-control AI[11]. In 2017, 75-375 million workers, accounting for 3–14% of the global workforce, might need to change occupations or upgrade skills by 2030, depending on AI adoption rates[12]. Moreover, as the landscape changes, it is essential for all workers to collaborate effectively with increasingly advanced machines and prepare to address the challenges of future employment[13]. Teachers, crucial to the education system, play a pivotal role in adapting to the evolving AI-centered landscape. Prioritizing educational quality necessitates technological acceptance and adjustment. Addressing teachers' anxiety, specifically AIA, becomes essential for effective AI integration in education[14].

The results from the group interviews suggest that Chinese university teachers generally have a high level of acceptance for generative AI tools. Additionally, middle-level administrators from various colleges indicate that their institutions were actively preparing for the declaration of AI-related art programs and curriculum development. Teachers also report active preparations for courses and platforms related to art and technology, such as digital sketching. According to the survey, the overwhelming majority of surveyed teachers indicated that they would offer AI-related courses(Fig. 4).

Fig. 4. You will design a completely AI-centric new course.

According to the group interviews, the type of institution where teachers work significantly influences their attitudes and applications of AI in art education. Teachers from art colleges tend to be more cautious and express concerns that AI technology may threaten the essence of art(Fig. 5). In contrast, teachers from comprehensive universities, especially those with backgrounds in interdisciplinary collaboration between Art and Engineering disciplines, are more inclined to embrace digital art and AI tools(Fig. 6). Although some of them think AI tools may improve teaching efficiency. The teacher from the China Central Academy of Fine Arts thinks art colleges lack technology-oriented student and faculty teams compared to comprehensive universities, making it challenging to establish art and technology courses and labs. The Tianjin Academy of Fine Arts teacher mentions that animation students require extensive making time, limiting the capacity to offer theoretical courses, including AI-related ones. Representatives from Xi'an Academy of Fine Arts worry about the impact of AI tools on traditional art education, feeling that the school faces the risk of potential discontinuation. They emphasize the need for collaboration among universities to address this issue and prevent professional art programs from becoming intangible cultural heritage.

Simultaneously, some teachers concern about how to integrate AI tools rather than being replaced by them. They aim to strike a balance between technical proficiency and aesthetic cultivation, leveraging the potential of AI tools while preserving the essence of art. Furthermore, some institutions highlight the need for collaboration and involvement of businesses to promote AI education. Some faculties suggest that interdisciplinary education is a path for AIAED. They propose building interdisciplinary course modules by focusing on multidisciplinary teachers within specialized departments, facilitating the development of cross-disciplinary courses.

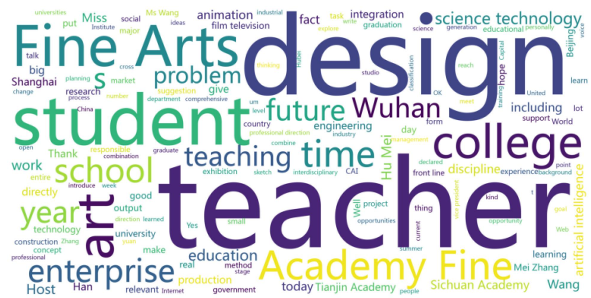

Fig. 5. Art college group interview word cloud.

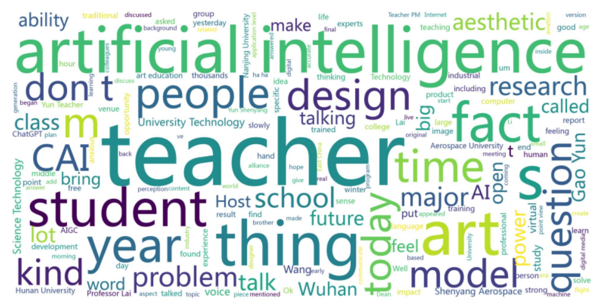

Fig. 6. Public university group interviews word cloud.

V. WILL AI REPLACE HUMAN TEACHERS AND ARTISTS?

The technologisation of education has had a deep impact on teachers and teaching[15]. Some interviewed teachers express their views on the potential and limitations of AI tools in art education. According to the survey, teachers generally believe that generative AI has application potential in art education while posing a threat to traditional art education(Fig. 7). They also indicate that the current state of digital art is still limited, and there is skepticism about whether AI-generated content can truly constitute digital art.

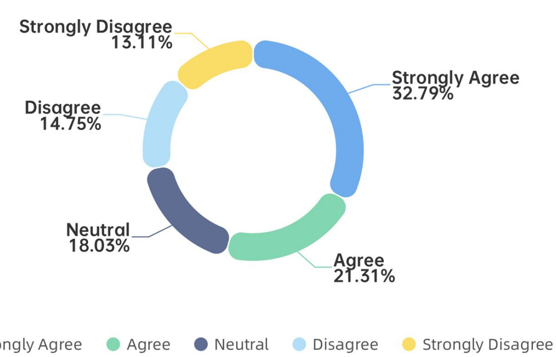

Fig. 7. You think some traditional teaching content or skills are no longer important because of introduction of AI tools.

A few teachers emphasize the importance of respecting the essence of art when using AI tools. They advocate for in-depth exploration of the impact of technology on art and call for finding an appropriate place for AI tools in art education. Others express confidence in the current state of AI art and encourage a focus on the aesthetic significance and value of digital art. They propose establishing a set of criteria for judging AI-generated art and creating a dedicated competition category for AIGC. However, the majority of teachers believe that current(AI produced) digital art isn't genuine AI art because it lacks the presence of digital genes.

In addition, some teachers think that AI excels in artistic presentation due to millennia of human accumulation. They believe in the future, artificial intelligence can continue to create things belonging to human aesthetics and cognition, allowing beauty to spread its wings freely. From the perspective of art colleges, there is no need for self-deprecation. Instead, they advocate a return to the essence of art and a deep exploration of what art is after technological changes. What truly defines digital art? Art, as a form of originality, is the release of the freest human soul, and this primal drive is eternal. In the AI era of art education, what students truly lack is artistic literacy, not the ability to operate AI tools. Computers do what they should, leaving what humans should do, which aligns with the findings of Chan's research[16].

Professor Zhou from Wuhan University of Technology proposes that in the development of art and design education, the emphasis has shifted from foundational design principles to software learning, and recently, with the impact of artificial intelligence, universities have begun contemplating how to embrace AI tools. Essentially, these are tools of modern learning, and educators do not need to be so nervous. Professor Zhou further suggests that various disciplines should gradually design exemplary courses that integrate AI tools. This can be achieved by enhancing teachers' understanding of the efficiency of AI tools through inter-collegiate exchanges.

There has been a longstanding expectation that AI would alleviate time and effort invested by teachers in the labor-intensive process of grading student assignments and assessments[17]. Fu believes that AI tools bring new possibilities to art education and he's already applied digital human cloning technology in his class. Educators can use AI to create personal digital twins, which can be used to answer students' questions, significantly reducing pressure on teachers for outside tutoring, thus freeing up time for research and other tasks. However, some teachers express concerns and anxieties about the potential threats of AI. They worry that AI may gradually replace human teachers in the future, even disrupting the field of art. Nevertheless, some educators assert that human teachers possess unique qualities that are irreplaceable. Professor Sun, Zhejiang University, and Professor Zhou, Wuhan University of Technology, both point out that they are contemplating how to ensure that teachers and students are not replaced by AI tools. Accordingly, they will adjust their talent development plans to address this issue.

VI. EQUITY OR INEQUITY?

The impact of AI tools on educational equity has sparked intense debate in the academic community. AI programs have raised significant concerns about academic integrity due to the sophisticated and high-quality nature of their outputs. Scholars believe that the high quality of texts generated by ChatGPT creates a higher risk that students will cheat on online exams[18]. In less than two months post-launch, academic reports suggest that up to one-fifth of students have been identified using AI programs for university assessments[19]. According to previous studies, participants were able to distinguish between human and AI-generated text with an accuracy of 50-52%, similar to the chance outcome of a coin flip, which means humans can no longer tell the difference[20]. In addition, due to these student behaviors, certain universities prohibited the use of ChatGPT, and some academics labeled such tools as a 'threat' and a 'plague' on education[21][22]. By conpariosn, in China, the prohibition of AI tools such as ChatGPT has hindered the widespread use of cutting-edge AI tools by university teachers and students. As a result, the adoption of AI tools in Chinese higher education is relatively low and college teachers tends to promote the spread of AI tools.

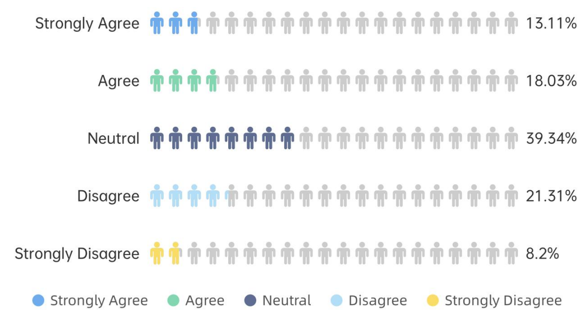

Fig. 8. You feel students' creations have largely changed after using AI.

During the interviews, the question of whether AI tools bring about fairness or unfairness receives widespread attention. Fu learns about AI tools when students discreetly used them for assignments. This teacher encourages such behavior, urging students to boldly embrace and experiment with new tools. In addition, a private college representative indicates that Chinese private colleges typically admit students with weaker hand-drawing skills, making it challenging for them to excel in Chinese discipline competitions. Besides, during the Artificial Intelligence Art Education Yangtze Forum in December 2023, Tian further discussed how AI tools are gradually leveling the fairness issue caused by uneven distribution of art education resources due to regional disparities[23]. In this context, the emergence of AI tools, to some extent, provides a 'fair' competitive opportunity for students with weaker drawing skills. As to the survey result, the number of teachers who agree, disagree, and remain undecided on whether students' creativity is enhanced by AI tools is roughly equal(Fig. 8).

However, Some interviewed teachers think school administrators and teachers have yet to provide reasonable solutions to address the issue of students using AI for cheating their graduation projects, and students using generative AI tools to 'cheat' may hinder the development of their professional skills. Additionally, teachers, like students, also face the question of whether they can maintain their boundaries in the face of AI(Fig. 9).

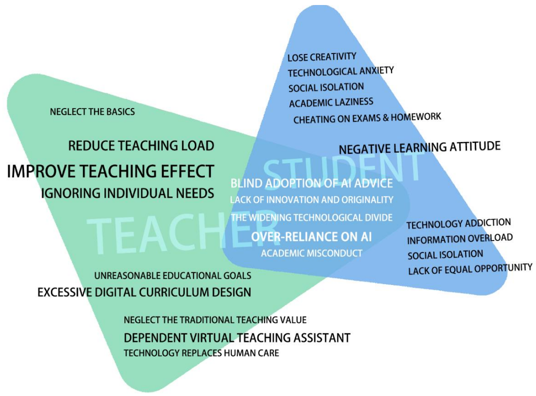

Fig. 9. AI temptations and potential pitfalls

Furthermore, there is an issue of educational resource inequality in the education sector. Furthermore, some teachers mention a culture of inertia within the education community, where senior educators expect younger ones to take on the responsibility of innovation and reform. Although scholars have defined the boundaries of AI cheating based on the use of AI tools in higher education[24], and have proposed relevant policy regulations[16], the journey ahead is still long.

VII. Meeting the Challenges of AIAED

In response to the challenges of AIAED, teachers have discussed and recommended various strategies. Teachers unanimously agree on the necessity of in-depth research to leverage AI tools for enhanced art education, emphasizing the urgency of teacher training. While some teachers have made efforts to incorporate AI tools, the majority lack sufficient knowledge. Although certain universities provide training on large AI models, most institutions lack plans to support teachers, potentially leading to inequalities among them. Consequently, there's increasing demand for teachers to gain digital skills for AI integration. Interdisciplinary learning is challenging, especially for older teachers.

AI art education requires not only deep educational knowledge but also robust computational power. However, high costs can exacerbate resource inequalities, resulting in educational disparities. To address growing demand for AI in education, suggestions include nonprofit training, standardized AI tool regulations, collaboration with super-computing centers, and university system restructuring. Educational resource inequality exists, with older educators expecting younger ones to lead innovation. Motivating teachers, changing outdated beliefs, and increasing student interaction are key for AIAED. Teachers should focus on rational research and tailor AI development to school needs. Frequent AIGC university exchanges and branch meetings are encouraged to boost participation.

VIII. Conclusion

The study aims to explore the attitudes of Chinese university art teachers towards integrating generative AI tools, investigating their acceptance, concerns, and preparations. Utilizing group interviews and questionnaires during the AIAED Alliance session, middle-level administrators and teachers from diverse colleges provided qualitative insights. Group interviews facilitated a nuanced understanding of diverse viewpoints. The research covered various educational institutions, ensuring a broad spectrum of perspectives, including comprehensive universities and art colleges.

This study delves into the multifaceted landscape of AI in higher art education, revealing nuanced perspectives among teachers. Teachers' acceptance of generative AI tools varies, with comprehensive universities showing readiness, while art colleges exercise caution. The study highlights the potential of AI in art education but underscores concerns about its impact on traditional pedagogy. Equity issues surface regarding resource inequalities, academic integrity, and cultural resistance within the education community. Teachers stress the urgency of training, interdisciplinary collaboration, and a cultural shift to harness AI's potential effectively. Recommendations include standardized AI tool usage, adapting professional structures, and fostering collaborative alliances.

This study acknowledges certain limitations. The sample size, primarily concentrated within the population of educators in animation and digital media programs, may limit the generalizability of findings. The reliance on qualitative methods, including group interviews, introduces subjectivity and potential information source bias. The study focused on current attitudes, and future research could explore how attitudes change with technological developments. Future research could delve deeper into the practical application of AI tools in higher education classrooms and comparing traditional and AI-assisted teaching methods could provide insights into enhancing teaching efficiency. Lastly, Exploring regional or cultural variations in the perspectives of art educators on AIGC can provide a more nuanced understanding of the subject.

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